

Credit Crunches with Recursive Preferences^{*}

Oliver de Groot[†]

Panagiotis Veneris[‡]

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Abstract

This paper studies the impact of recursive utility on household behavior in an otherwise standard model of incomplete markets and heterogeneous agents. By incorporating Epstein-Zin preferences, we demonstrate that household and aggregate responses to shocks are significantly dampened compared to a model with time separable (CRRA) utility. This is because recursive preferences imply that risk aversion and intertemporal substitution are distinct structural parameters, and allowing the former to take on large values we are able to generate substantial swings in consumer behavior. We further show that this finding is overturned when considering a different scenario of credit tightening, such as an increase in credit spreads, highlighting the broader implications of recursive preferences for macroeconomic dynamics.

JEL Codes: D52, D80, E32, E44, G52

Keywords: Heterogeneous Agents; Incomplete Markets; Recursive Utility; Precautionary Savings

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[†]Corresponding author: University of Liverpool Management School & CEPR, Chatham Street, Liverpool, L69 7ZH, UK. Email: oliverdegroot@gmail.com

[‡]University of Liverpool Management School, Chatham Street, Liverpool, L69 7ZH, UK. Email: Panagiotis.Veneris@liverpool.ac.uk

1 INTRODUCTION

In this paper we study the effects of recursive utility on the response of the economy to an exogenous contraction in borrowing capacity. Extending the [Guerrieri and Lorenzoni \(2017\)](#) incomplete markets model to incorporate [Epstein and Zin \(1989\)](#) preferences, we evaluate households' optimal policies and their propagation to aggregate economic activity when credit gets tighter.

Our findings are twofold. First, we show that the model with recursive utility implies a smoother amplification to a tightening in the borrowing constraint relative to the economy with time separable preferences. This is in contrast to the conventional wisdom, whereby greater future uncertainty associated with a financial crisis tends to generate stronger idiosyncratic and aggregate movements in macroeconomic variables. With Epstein-Zin preferences, higher risk aversion makes unconstrained households react to the borrowing shock by accumulating more precautionary savings. Therefore, the adjustment through the intensive margin calls for a stronger amplification. On the extensive margin, higher risk aversion implies that fewer households are close to or against the credit constraint in the first place. As such, the deleveraging wave becomes more moderate. In the aggregate, the extensive margin dominates and the response of aggregate variables gets muted. Second, we show that a uniform increase in credit spreads — i.e. a symmetric rise in the cost of borrowing — overturns the previous result. Now, households with recursive preferences tend to overreact to the shock relative to the time separable case. Intuitively, higher credit spreads operate in tandem with the stronger precautionary savings motive in the Epstein-Zin economy, inducing more pronounced individual and aggregate responses.

In the model, a unit measure of households with Epstein-Zin preferences consume, trade in the market for government bonds, and — if employed — produce a final good. Households differ with respect to initial asset holdings and employment status. All households face the same (exogenous) borrowing constraint but receive idiosyncratic draws to labor productivity (employment). In that sense, markets are incomplete and the government bond is the only traded asset in the economy. The borrowing constraint occasionally binds for some households, conditional on the initial asset position and the (contemporaneous) income realization. This, along with incomplete insurance, drives the internal dynamics of households' decisions, and makes the distributional impact of shocks matter for the aggregates ([Heathcote et al., 2009](#)).

Conceptually, incorporating recursive utility allows us to isolate the role of attitudes toward intertemporal substitution from risk aversion in shaping households' responses to financial shocks. During a credit tightening, the perceived uncertainty about future income and borrowing shocks rises sharply. Standard CRRA preferences compress these two dimensions into a single curvature parameter, thereby masking the distinct responses to risk and timing. Recursive utility, by contrast, enables a more nuanced characterization of precautionary behavior and intertemporal substitution, which are central to understanding household deleveraging and the ramification of credit shocks. From an empirical standpoint, standard incomplete markets models with CRRA utility often struggle to match the observed strength and persistence of aggregate responses to credit contractions. The Great Recession, for example, featured a sharp rise in perceived uncertainty, elevated precautionary savings, and a prolonged period of low real interest rates. These patterns suggest that risk attitudes and intertemporal substitution operate through separate channels — a feature captured naturally by recursive preferences. Embedding Epstein–Zin utility in a heterogeneous-agent environment therefore provides a more realistic laboratory to study the joint dynamics of consumption, savings, and (potentially) asset prices in times of financial distress.

Literature Our paper relates to at least three strands in the literature. First, it builds on the body of research that studies the importance of individual heterogeneity in quantitative incomplete markets models — a long-lasting theme in the macroeconomics research agenda, which received little attention until the wake of the financial crisis. Early papers that embrace heterogeneity are the seminal works of [Bewley \(1977\)](#), [Imrohoroglu \(1989\)](#), [Huggett \(1993\)](#) and [Aiyagari \(1994\)](#). [Imrohoroglu \(1989\)](#) quantifies the welfare cost of business cycles in a model with incomplete insurance and stochastic employment draws. Her findings suggest a much larger cost (about six times) compared to the benchmark with full insurance, when a no-borrowing constraint is imposed. In an endowment model, [Huggett \(1993\)](#) explores the effect of ex-post heterogeneity and partial insurance on the risk-free rate, and finds this combination to generate a more realistic equilibrium real interest rate, lower than what representative-agent complete markets models could predict. Extending [Huggett \(1993\)](#) in a production economy, [Aiyagari \(1994\)](#) studies the quantitative implications of individual income uncertainty for aggregate savings (and the

distributions of income and wealth) and shows that for plausible values of risk aversion idiosyncratic risk only has a marginal effect. Recently, [Bayer et al. \(2019\)](#) analyze the impact of heightened income risk on aggregate activity, and its distributional effects when markets are incomplete. The analysis marks income risk as a crucial factor for the intensity of the Great Recession. Complete markets models that mute the role of idiosyncratic uncertainty cannot capture the distributional impact of shocks, delivering predictions that are orthogonal to empirical findings. For this reason, we follow the incomplete markets paradigm by assuming that households can only partially insure against idiosyncratic risk, in a consumption-savings model with household heterogeneity. Our contribution is to enrich this line of research by embedding recursive utility.

Second, our paper spans the pool of work on dynamic macroeconomic and macro-finance models with [Epstein and Zin \(1989\)](#) preferences. In general, recursive preferences are particularly relevant for studies on the intersection of macroeconomics and finance, for their ability to predict realistic asset prices and equity risk premia. The most representative study that popularized the practicality of recursive preferences is the seminal work of [Bansal and Yaron \(2004\)](#). Using a model with Epstein-Zin preferences and long-run consumption risk, they match empirical estimates of the equity premium and the risk-free rate, offering a potential resolution to the corresponding puzzles. [Rudebusch and Swanson \(2012\)](#) fit stylized business cycle and asset pricing moments augmenting Epstein-Zin preferences and long-run inflation risk in a vanilla New Keynesian setup. [Pidkuyko et al. \(2019\)](#) add durable consumption in a standard [Lucas \(1978\)](#) tree economy with recursive utility to address the puzzle of excessive timing and risk premia implied by long-run risk models. [Nezafat and Slavík \(2021\)](#) build a heterogeneous-agent, production-based asset pricing economy with productivity and liquidity shocks to reconcile business cycle fluctuations and the levels of the equity premium and the risk-free rate. [de Groot et al. \(2022\)](#) estimate a sequence of endowment asset pricing models with long-run cash-flow risk and stochastic volatility to highlight the importance of the proper incorporation of time-preference shocks in the Epstein-Zin utility for the success of valuation risk in predicting asset prices. Most of these papers — with a notable exception of [Nezafat and Slavík \(2021\)](#) — rely on models with a representative agent and complete markets to bridge the gap between asset pricing predictions and business cycle frequencies. Our paper extends this literature, and complements [Nezafat and Slavík \(2021\)](#), by examining

the dynamics of a credit crunch in a unified framework featuring household heterogeneity and uninsurable idiosyncratic risk. The main deviation from [Nezafat and Slavík \(2021\)](#) is that our model does not admit explicit aggregation.

Third, our paper connects with a growing body of work that stresses the importance of precautionary savings in business cycle models. For example, [Challe and Ragot \(2016\)](#) build a heterogeneous agent economy in which precautionary savings arise as a cushion against countercyclical unemployment risk. They show that the response of aggregate consumption is amplified through the precautionary savings channel, propagating business cycle fluctuations. In a similar manner, [Bayer et al. \(2019\)](#) gauge the contribution of precautionary savings and portfolio rebalancing in aggregate activity. Similar to the aforementioned papers, central in our model is the role of precautionary savings for the amplification of exogenous disturbances. However, contrary to [Challe and Ragot \(2016\)](#) which makes simplifying assumptions to get exact aggregation, in our model the cross-sectional wealth distribution matters for buffer stock savings as it induces unconstrained households to save more in response to a tightening in borrowing capacity, leading to a sharper decline in aggregate consumption and the equilibrium interest rate.

The remainder of the paper proceeds as follows. [Section 2](#) lays out the model and discusses the preference specification. [Section 3](#) presents the steady states of the model, and the key takeaways from the credit crunch experiment. [Section 4](#) contrasts the findings from the expected-utility case with those under the Epstein-Zin specification. [Section 5](#) outlines the model with credit spreads. [Section 6](#) concludes.

2 MODEL

In modeling the household sector, we closely follow the approach in [Guerrieri and Lorenzoni \(2017\)](#). Time is discrete and the horizon is infinite. Uncertainty in this economy comes from exogenous idiosyncratic shocks to labor productivity drawn randomly from its ergodic distribution. The economy is populated by a continuum of households that make state-contingent plans about consumption and labor supply, pay labor income taxes, and trade on a one-period, non-contingent, risk-less bond to smooth the effects of idiosyncratic risk on consumption. Households differ each period with respect to income and their initial asset position. The absence of a full set of state-contingent Arrow securities implies that the only avenue of insurance against negative income surprises is through bond accumulation. In that sense, our model features exogenous incomplete asset markets. For each household i , $c_i \geq 0$, and $n_i \in [0,1]$ are consumption and labor supply, respectively, and $\tilde{\tau}_i$ are taxes on labor income.

Every household produces consumption goods using a linear technology of the form

$$y_i = \theta_i n_i, \tag{1}$$

where θ_i denotes the idiosyncratic labor productivity shock that household i faces in period t . The shock is positive and follows an AR(1) process in logs, approximated by an S -state Markov chain on the space $\{\theta^1, \dots, \theta^S\}$.¹ We also allow for unemployment by assuming that households hit by an income shock $\theta^1 = 0$ to be unemployed. Income realizations span the interval $\{\theta^1, \dots, \theta^S\}$. Household i 's budget constraint is given by

$$qb'_i + c_i \leq b_i + y_i - \tilde{\tau}_i, \tag{2}$$

where b_i are bond holdings traded at price q , b'_i denotes next period bonds, and $\tilde{\tau}_i$ are tax payments. The tax structure is the following - while every household pays a lump-sum tax τ , the unemployed receive a transfer ν in a lump-sum fashion, which can be interpreted as an unemployment benefit. Thus, $\tilde{\tau}_i = \tau$ for employed ($\theta_i > 0$), and $\tilde{\tau}_i = \tau - \nu$ for unemployed ($\theta_i = 0$). Households are constrained to borrow up to an exogenous limit ϕ ,

¹All realizations of θ in this interval are positive and in ascending order.

and hence bond holdings each period must satisfy a borrowing constraint of the form²

$$b'_i \geq -\phi. \quad (3)$$

The government levies lump-sum taxes on households and issues non-contingent debt to finance an exogenous stream of government expenditures, consisting of transfers ν given to a fraction $u = Pr(\theta_i = \theta^1)$ of unemployed in the population, and outstanding debt obligations. More formally, the government budget constraint takes the form

$$B + u\nu = qB' + \tau. \quad (4)$$

Under the assumption of a fixed supply of government bonds B and unemployment benefits ν , the government can always balance the budget rule by adjusting the tax schedule τ .

We close the model by making two simplifying assumptions. First, we assume that in this economy the only supply of liquid assets B , comes either from the household sector or from the government.³ Assets play a crucial role in households' behavior, since they act as shock absorbers and the only avenue of insurance against current and future low-endowment realizations. Finally, we assume that the rate at which households borrow is the same as the rate at which they lend, that is we abstract from interest rate wedges.⁴

2.1 Epstein-Zin preferences

Household preferences follow the recursive structure of [Kreps and Porteus \(1978\)](#). We focus on the isoelastic preferences of [Epstein and Zin \(1989\)](#) and [Weil \(1990\)](#), which generalize expected utility time-separable preferences with endogenous labor supply, and can be written recursively as a Bellman equation

$$\tilde{V}(b_i, \theta_i) = \left[u(c_i, 1 - n_i) + \beta \left(E\tilde{V}(b'_i, \theta'_i)^{\tilde{\alpha}} \right)^{\frac{\rho}{\tilde{\alpha}}} \right]^{\frac{1}{\rho}}. \quad (5)$$

²The borrowing constraint is assumed to be always tighter than the natural borrowing limit, and occasionally binds in equilibrium.

³We implicitly normalize net assets held by the household sector to equal B .

⁴We relax this assumption in [Section 5](#).

Households derive utility from consumption and leisure, and a certainty equivalent of continuation utility. If we let $V = \tilde{V}^\rho$ and $\alpha = 1 - \tilde{\alpha}/\rho$, then equation (5) corresponds to

$$V(b_i, \theta_i) = u(c_i, 1 - n_i) + \beta (EV(b'_i, \theta'_i)^{1-\alpha})^{\frac{1}{1-\alpha}}. \quad (6)$$

$u(c_i, 1 - n_i)$ is period utility from the stream of consumption and leisure, $\beta \in (0,1)$ is the discount factor, $E(\cdot)$ is the expectations operator conditional on period t information, $\mu = (EV(b'_i, \theta'_i)^{1-\alpha})^{\frac{1}{1-\alpha}}$ is a certainty equivalent of next period's continuation utility, and α is the Epstein-Zin parameter. The Epstein-Zin parameter is a linear combination of the risk aversion parameter $\tilde{\alpha}$ and the elasticity of intertemporal substitution $1/\rho$. The certainty equivalent operator translates random future continuation utility into units of the composite good comprising consumption and leisure.

For the period utility we assume that it is additive and isoelastic in consumption and leisure, such that

$$u(c_i, 1 - n_i) = \frac{c_i^{1-\gamma}}{1-\gamma} + \psi \frac{(1 - n_i)^{1-\eta}}{1-\eta}. \quad (7)$$

$u(c_i, 1 - n_i)$ is twice continuously differentiable, strictly increasing and concave in consumption, and strictly decreasing and convex in labor supply. The usual non-negativity constraint on consumption, $c_i \geq 0$, and the feasibility constraint on labor, $n_i \in [0,1]$, apply.

If $u \geq 0$ everywhere, then there exists a value function $V \geq 0$ that solves (6), and the utility recursion becomes⁵

$$V(b_i, \theta_i) = \frac{c_i^{1-\gamma}}{1-\gamma} + \psi \frac{(1 - n_i)^{1-\eta}}{1-\eta} + \beta (EV(b'_i, \theta'_i)^{1-\alpha})^{\frac{1}{1-\alpha}}. \quad (8)$$

Parameter ψ measures how much households value leisure, and η stands for the inverse Frisch elasticity of labor supply to changes in the wage rate. The Epstein-Zin composite parameter is defined as $\alpha \equiv 1 - RA/\gamma$, and $1/\gamma$ governs the (constant) elasticity of intertemporal substitution between the composite good and the certainty equivalent of continuation utility.⁶

The difference between the Epstein-Zin and a standard expected utility setting is that in the former, the value function is twisted and untwisted by $1 - \alpha$, and the coefficient

⁵Theorem 3.1 in [Epstein and Zin \(1989\)](#).

⁶There is an equivalence between the definition of the Epstein-Zin parameter in equations (8) and (6), with $RA \equiv \tilde{\alpha}$, and $1/\gamma \equiv 1/\rho$.

of risk aversion is amplified or attenuated by the curvature parameter α . However, the main feature of Epstein-Zin utility is that it separates risk aversion from the elasticity of intertemporal substitution (EIS). These two parameters are now allowed to vary independently of each other. While EIS in (8) is defined symmetrically to the expected utility counterpart, risk aversion becomes linear in the EIS and the Epstein-Zin parameter. Therefore, our setup provides an extra degree of freedom as we can vary risk aversion without affecting the elasticity of intertemporal substitution.⁷

Finally, if $u \leq 0$ everywhere, then to avoid complex numbers we let $V \leq 0$ and rewrite the recursion (8) as

$$V(b_i, \theta_i) = \frac{c_i^{1-\gamma}}{1-\gamma} + \psi \frac{(1-n_i)^{1-\eta}}{1-\eta} - \beta \left(E(-V(b'_i, \theta'_i))^{1-\alpha} \right)^{\frac{1}{1-\alpha}}. \quad (9)$$

Recall that a household with recursive utility as in (6) is averse to volatility in future utility — i.e. prefers early resolution of uncertainty — if and only if $RA > \gamma$. In contrast, the household accommodates volatility in future consumption — i.e. prefers late resolution of uncertainty — if and only if $\gamma > RA$. Here we make the assumption that $RA > \gamma$, which implies that recursive utility increases the curvature with regard to future risks, making agents strictly prefer early resolution of uncertainty.

2.2 Equilibrium

Household's Problem Households choose a stream of consumption and labor supply, $\{c, n\}_{t=0}^{t=\infty}$, to maximize their lifetime utility, $V(c, n)$, subject to the budget constraint (2), the non-negativity constraint on consumption, $c \geq 0$, and the feasibility constraint for labor, $n \in [0, 1]$. Using the budget constraint, we can express the choice variables as functions of the state variables, that is, $C(b, \theta)$ and $N(b, \theta)$.

Bellman Equation We assume that household i 's value function exists and satisfies the Bellman equation

$$V(b_i, \theta_i) \equiv \max_{\{c_i, n_i, b'_i\}} \left[u(c_i, 1 - n_i) + \beta \left(EV(b'_i, \theta'_i)^{1-\alpha} \right)^{\frac{1}{1-\alpha}} \right], \quad (10)$$

⁷In general, this property is critical for the proper study of asset returns, and the resolution of some long-standing asset puzzles.

subject to

$$c_i \geq 0, \quad n_i \in [0, 1], \quad (11)$$

and the household's budget constraint (2). The optimal choices of consumption and labor supply are summarized in two equations.⁸ The intertemporal optimality condition (Euler for consumption)

$$u_{c_i}(c_i, 1 - n_i) \geq \beta(1 + r) [EV(b'_i, \theta'_i)^{1-\alpha}]^{\frac{\alpha}{1-\alpha}} E [V(b'_i, \theta'_i)^{-\alpha} u_{c'_i}(c'_i, 1 - n'_i)], \quad (12)$$

captures the trade-off households face between consumption today and in the future. Kuhn-Tucker conditions ensure that as long as the debt constraint $b'_i \geq -\phi$ does not bind, this equation holds with equality. The intratemporal optimality condition (Euler for labor supply) is given by

$$-\frac{u_n(c_i, 1 - n_i)}{u_c(c_i, 1 - n_i)} \geq w, \quad (13)$$

and equates the marginal rate of substitution between consumption and leisure with the real wage. It holds with equality when $n_i \geq 0$.

Definition 1. A recursive competitive equilibrium in this economy is a sequence of value functions $\{V\}: \mathbb{R}_+^2 \rightarrow \mathbb{R}$, decision rules $\{C, N\}: \mathbb{R}_+^2 \rightarrow \mathbb{R}$, interest rates $\{r\}: \mathbb{R} \rightarrow \mathbb{R}$, tax rates $\{\tau\}: \mathbb{R}_+ \rightarrow \mathbb{R}_+$, and the (joint) distribution of bond holdings and productivity levels $\{\Psi\}: \mathbb{R}^2 \rightarrow \mathbb{R}_+$, such that (i) given the interest rate $\{r\}$ and the tax rate $\{\tau\}$, the value function $V(b, \theta)$ solves the Bellman equation (10), and $\{C(b, \theta), N(b, \theta)\}$ are the corresponding optimal decision rules, (ii) the (joint) distribution of bond holdings and productivity levels Ψ is consistent with the households' optimal consumption and labor supply decisions, (iii) the tax rate equals government expenditures

$$\tau = \nu u + \frac{rB}{1 + r},$$

and the asset (bond) market clears

$$\int b d\Psi(b, \theta) = B.$$

⁸See Appendix A for analytical derivations.

2.3 Calibration

Table 1. Baseline Calibration

Parameter	Value	Description	Target/Source
β	0.9749	Discount factor	$r = 2.5\%$
γ	4	Inverse elasticity of intertemporal substitution	
RA	15	Coefficient of relative risk aversion	
η	1.5	Curvature of utility from leisure	Average Frisch elasticity = 1
ψ	15.57	Weight on utility from leisure	Nekarda and Ramey (2020)
ρ	0.967	Productivity shock persistence	Floden and Lindé (2001)
σ_ϵ	0.017	Productivity shock variance	Floden and Lindé (2001)
$\pi_{e,u}$	0.057	Transition to unemployment	Shimer (2005)
$\pi_{u,e}$	0.882	Transition to employment	Shimer (2005)
B	1.60	Bond supply	Liquid assets
$\phi/4Y$	1.102	Borrowing constraint	Total gross debt
ν	0.167	Unemployment benefit	40% of average labor income

The calibration of the model is based on [Guerrieri and Lorenzoni \(2017\)](#), and utilizes quarterly frequency data to match some salient features of the US economy. The baseline parameters are shown in Table 1. The discount factor is calibrated to $\beta = 0.9749$, such that it yields an annual net real interest rate of 2.5%.⁹ For the elasticity of intertemporal substitution, typical values in the literature vary widely, mostly ranging from 0 to 1, and in some instances to 2.¹⁰ In our baseline calibration we set $1/\gamma = 1/4$, a value upon which we experiment later. The parameter that governs the inverse Frisch elasticity is calibrated to $\eta = 1.5$, to yield a unitary elasticity of labor supply that is within the range of micro estimates ([Chetty et al., 2011](#)), and widely used in macroeconomic studies.¹¹ The weight on the utility from leisure is calibrated to $\psi = 15.57$ to target an average of hours worked of 40% of the time endowment of employed workers, as empirically estimated in [Nekarda](#)

⁹Note that here we follow an unconventional approach. Instead of fixing the value of the discount factor and search for the steady state interest rate consistent with our β , we set the interest rate and search for the β that pins down that interest rate.

¹⁰[Clark and Summers \(1982\)](#) estimate the EIS to be close to 1, and [Hall \(1988\)](#) finds that EIS ranges from 0 to 0.2. [Attanasio and Weber \(1993\)](#) and [Attanasio and Weber \(1995\)](#) find an elasticity of substitution equal to 0.8. Recent studies, such as [Bansal et al. \(2016\)](#), [Best et al. \(2020\)](#), and [Martinez et al. \(2021\)](#) report estimates of 2.18, 0.1, and 0.5, respectively.

¹¹See among others [Justiniano et al. \(2015\)](#), [Bigio and Schneider \(2017\)](#), [Debortoli and Galí \(2017\)](#), [Perri and Quadrini \(2018\)](#), [Karantounias \(2018\)](#), [Boppart et al. \(2018\)](#), [Kaplan et al. \(2018\)](#), [Bayer et al. \(2019\)](#), [Cui and Sterk \(2021\)](#), [Debortoli and Galí \(2022\)](#).

and Ramey (2020). Using data on wages and hours worked at the micro level, Floden and Lindé (2001) estimate the stochastic wage process for the U.S, and show that the persistent component displays a degree of autocorrelation equal to 0.91, and a conditional variance of 0.0426. Those values are also found in McKay et al. (2016) and Auclert (2019). Therefore, the degree of persistence and the variance of our logarithmic AR(1) process, ρ and σ_ϵ , are set accordingly. Shimer (2005), using unemployment-duration data adjusted by the BLS, estimates the job finding and separation rates at business cycle frequencies. Thus, the transition probabilities from employment to unemployment, and from unemployment to employment, are calibrated to $\pi_{e,u} = 0.057$ and $\pi_{u,e} = 0.882$, respectively. B reflects the sum of bond holdings held by the household sector, and is set to $B = 1.60$ to match the (liquid) assets to GDP ratio of 1.78 in the US in 2006. The borrowing limit — as a share of total annual output — is calibrated to $(\phi/4Y) = 1.102$, such that it yields a 18% (annualized) consumer debt-to-GDP ratio in the initial steady state. The unemployment benefit is calibrated to $\nu = 0.167$, to account for 40% of average labor income (Shimer, 2005).

There is little consensus with regards the range of plausible values for the coefficient of relative risk aversion, with the choice being mainly dependent on the model economy.¹² For instance, representative agent DSGE models with model-consistent expectations and low uncertainty about the economic environment (rational expectations hypothesis), need relatively higher values of risk aversion to match asset moments in the data (Rudebusch and Swanson, 2012). In contrast, long-run risk models largely respect the upper bound of ten (Epstein et al., 2014), as rationalized by Mehra and Prescott (1985). In our Epstein-Zin setup, idiosyncratic income uncertainty entails sufficient consumption risk, which allows us to set the coefficient of relative risk aversion to $RA = 15$.

¹²In their long-run risk model, Bansal and Yaron (2004) set the coefficient of relative risk aversion equal to 7.5. Piazzesi et al. (2006) fix it to 57, Rudebusch and Swanson (2012) and Van Binsbergen et al. (2012) use values of 75 and 80, respectively, while Karantounias (2018) uses a value of 10 for his benchmark calibration.

3 QUANTITATIVE ANALYSIS

In this section we (i) compute the steady state of the model, and (ii) quantitatively analyze the response of the economy to an exogenous tightening of the borrowing constraint.

3.1 Steady state

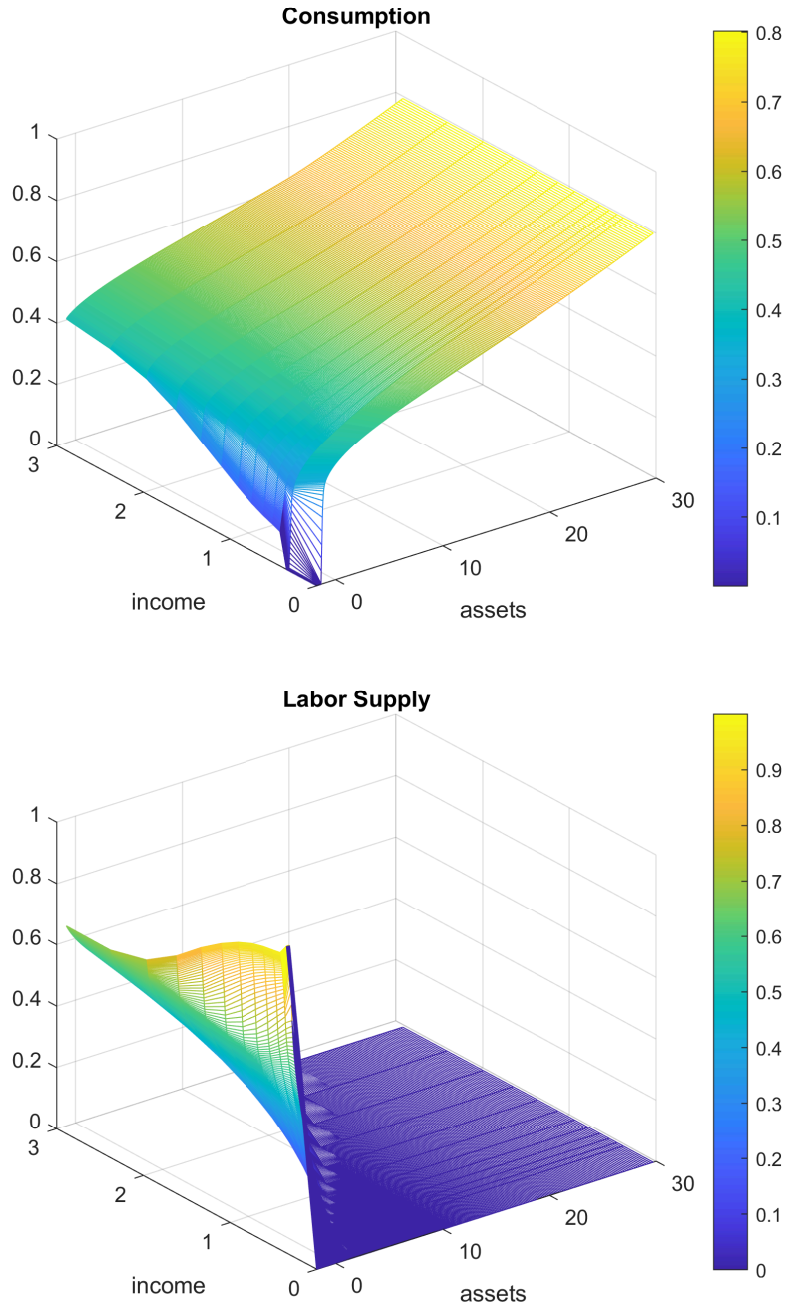
Figure 1 sketches the optimal decisions of households, with respect to consumption and labor supply, as a function of the initial levels of bond holdings and productivity, from the lowest productivity level θ^1 (unemployment state) to the highest productivity level θ^{13} . This is to uncover the different dynamics at play at different regions of the wealth distribution.

Two things are worth mentioning. First, there is much heterogeneity in the consumption and labor supply responses between low wealth and wealthy households, that is households at the left tail and the right tail of the wealth distribution, respectively. With regard to consumption, for low levels of bond holdings, the consumption function displays high concavity, while for high levels of bond holdings consumption is almost linear in wealth. The concavity of the consumption function came into the spotlight in the seminal work of [Zeldes \(1989\)](#) who argued that contrary to the case without income uncertainty in which the consumption rule is linear, the presence of uncertainty makes the consumption function concave in wealth. This is also summarized in [Carroll and Kimball \(1996\)](#) who prove that when income risk is introduced, the consumption of low-wealth households tends to be highly sensitive to changes in wealth or transitory income, while for wealthy households the opposite is true.^{13,14} When it comes to labor supply, our benchmark calibration yields a convex labor supply function, that is an inverse relationship between asset holdings and labor supply. This is intuitive, as households at the lower end of the distribution work harder to increase their labor income and better insure against negative future shocks, indicative of strong income effects that prevail at low levels of wealth. The concavity of the consumption function along with the convexity of labor supply are key ingredients for the determination of households' consumption and

¹³Its origin dates back to Keynes and his book the 'General Theory of Employment, Interest, and Money'.

¹⁴This is in contrast to [Kimball \(1990\)](#) who shows that the marginal propensity to consume alternates for given levels of consumption but not of wealth.

Figure 1. Policy functions | Epstein-Zin



NOTES: Consumption and labor supply decision rules under Epstein-Zin preferences. Variables are expressed in levels.

savings decisions, and hence for the behavior of the wealth distribution.

Second, in a recent paper [Wang et al. \(2016\)](#) show that the concavity of the consumption function is directly related to households' efficacy to self-insure against idiosyncratic risk, with better-insured households experiencing lower consumption volatility. The intuition is that at high levels of asset holdings, the materialization of the shock does not significantly

alter the consumption habits of households, given these households have the capacity to utilize their existing savings to hold on to a smooth consumption path. Conversely, the savings of households with low wealth are inadequate to keep their consumption levels unchanged, and hence the shock is partially absorbed by a downward adjustment in consumption. In that sense, consumption at low levels of wealth becomes more volatile and the marginal propensity to consume (*MPC*) increases. This is in line with our findings, and highlights the importance of the interplay between idiosyncratic risk and the occasionally binding borrowing constraint in understanding the behavior of the *MPC*, and the subsequent impact of the credit crunch. Quantitatively, our model delivers an aggregate (i.e. average) marginal propensity to consume of 0.019 in the initial steady state.

3.2 Credit crunch

In this subsection we describe our baseline experiment. The economy is in steady state in period $t = 0$, and the maximum amount of borrowing — as a fraction of (annualized) total output — allowed for households is limited to $(\phi/4Y) = 1.102$. In period $t = 1$ the economy is hit by an exogenous and unexpected one-time shock to the borrowing limit, which is assumed to be permanent in nature and to gradually reduce households' borrowing capacity to ϕ' (Guerrieri and Lorenzoni, 2017). That is, we allow the transition path from ϕ to ϕ' to be smooth, by letting ϕ periodically tighten by an amount equal to $\Delta\phi$. The value of $\Delta\phi$ is chosen such that households meet the new constraint after six periods. Thus, the evolution of household debt takes the general form

$$\phi_t = \max\{\phi', \phi - \Delta\phi \cdot t\}, \quad t \geq 1.$$

The aggregate shock hits symmetrically every household, and is calibrated to replicate a fall in the debt-to-GDP ratio of ten percentage points.¹⁵ This implies $(\phi'/4Y) = 0.600$.¹⁶

The reason for relying on a slow adjustment of the borrowing constraint is twofold. First, we model bonds as assets traded quarterly. In reality, debt maturities are much longer than a single quarter. Second, if we assume that households should meet the constraint in such a short time, this would require a sufficiently aggressive degree of

¹⁵This is similar to the decline in the U.S debt-to-GDP ratio in the apex of the Great Financial Crisis.

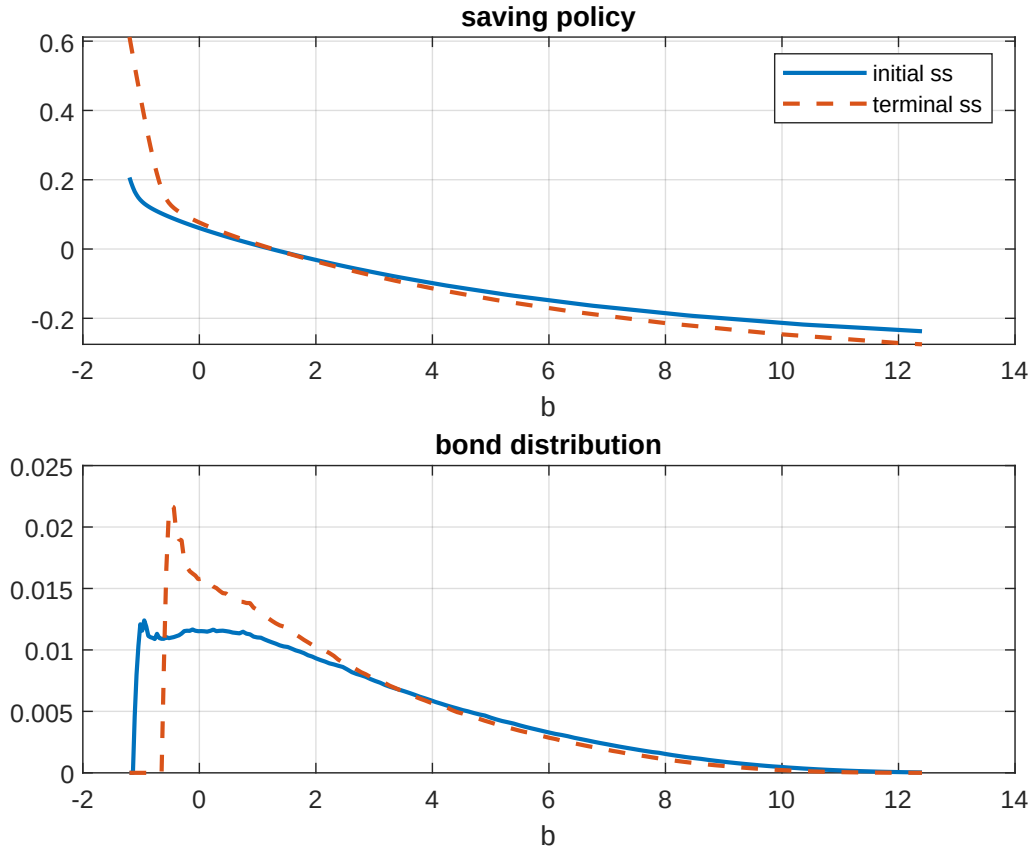
¹⁶We model the tightening in the borrowing limit as a non mean-reverting MIT shock.

adjustment — especially on the part of households closer to the constraint — which could induce some of them to default. To avoid all these unintended consequences, we let households gradually shift to the new lower borrowing limit. An important side effect of the assumption of a gradual transition to the tighter limit is the nontrivial implications for the adjustment path, with households' reaction being entirely dependent on their initial asset position.

3.2.1 Implications for savings and the distribution of assets

Figure 2 contrasts the demand for savings and the distribution of assets in the initial steady state with their counterparts in the terminal steady state.¹⁷

Figure 2. Savings and bond distribution | Epstein-Zin



We start with the first panel of figure 2 and the policy functions for savings. Our initial calibration delivers convex savings functions. The convexity of the savings functions

¹⁷The savings policy is determined by the ratio $(b' - b)/(4Y)$. More precisely, b' represents optimal savings averaged across employment states, computed as the product of the mass of households at each employment state and the optimal (individual) savings b'_i , as a function of the initial level of bond holdings b_i .

suggests that households at the left end of the distribution (low-wealth) want to increase their savings to move away from the borrowing limit. Comparing the behavior before and after the shock, we observe that following a credit tightening, low-wealth households increase their savings whereas those with high initial asset-holdings (wealthy) act in the opposite direction, cutting back on their savings in favor of a higher consumption path. The implications of the savings behavior manifest in the second panel, and the ergodic distribution of bonds. In particular, following a credit tightening the ergodic bond distribution concentrates to the right. However, since the aggregate supply of bonds is exogenously fixed, this shift should not be interpreted as an increase in the aggregate amount of savings, but rather a rebalancing of households' balance-sheets. More precisely households up against or close to the constraint increase asset demand, and those at the right tail of the asset distribution decumulate assets. In that sense, the aggregate level of savings is preserved across steady states, and both distributions display the same averages.

3.2.2 *Transitional Dynamics*

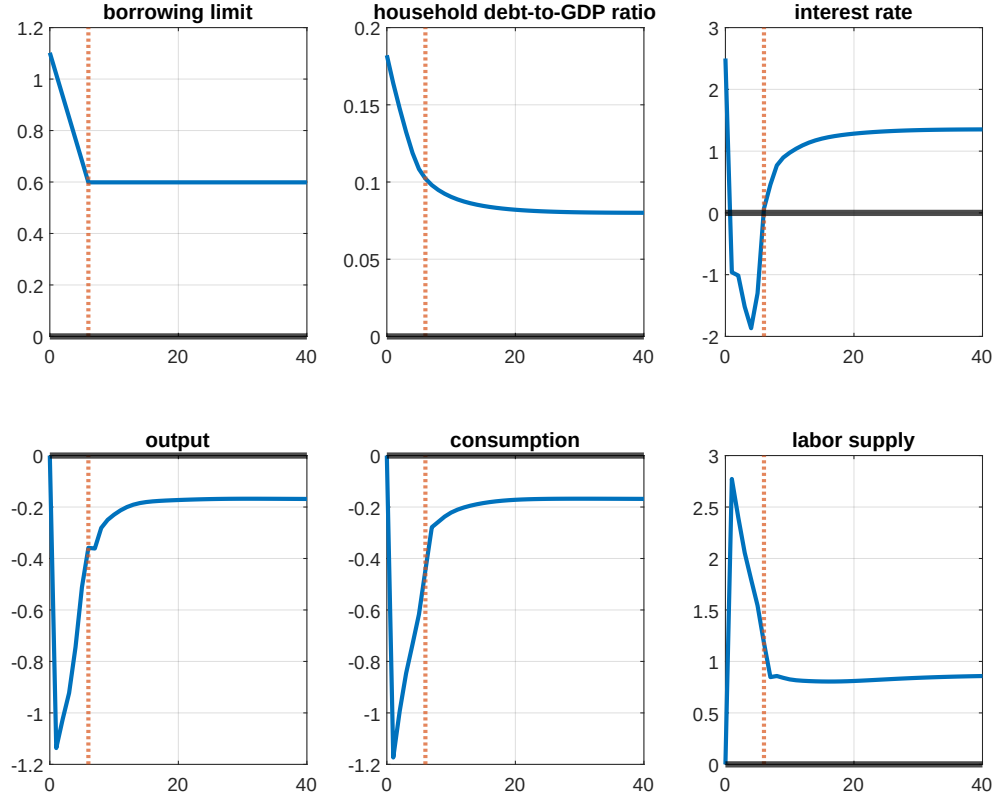
The aggregate behavior of households can be explained in terms of the consumption and labor supply functions, and the occasionally binding borrowing constraint.

When the aggregate shock hits, households who are now debt constrained, respond by sharply decreasing their consumption ($MPC_i \times \Delta\phi$),¹⁸ and increasing their labor supply in order to meet the now lower borrowing limit. This also holds for unconstrained households due to strong precautionary saving motives, and is the outcome of the partial equilibrium response to the credit tightening.

However, since our analysis is in a general equilibrium context, the endogenous response of prices, and in particular that of the interest rate, is pivotal for the precise inference about households' behavior, and the observed dynamics of savings and the bond distribution. Figure 3 shows the full transition path following the exogenous shock to the credit limit.

¹⁸For a formal proof see the appendix in [Guerrieri and Lorenzoni \(2017\)](#). Aggregation across households delivers an aggregate partial equilibrium consumption response given by $\overline{MPC} \times \overline{\Delta\phi} + Cov(MPC_i, \Delta\phi)$, where $\overline{MPC}$ is the average MPC in the economy, $\overline{\Delta\phi}$ the average shift in wealth, and the covariance term captures the extent to which household i is prone to the wealth redistribution succeeding the shock ([Auclert, 2019](#)).

Figure 3. Impulse response functions | Epstein-Zin

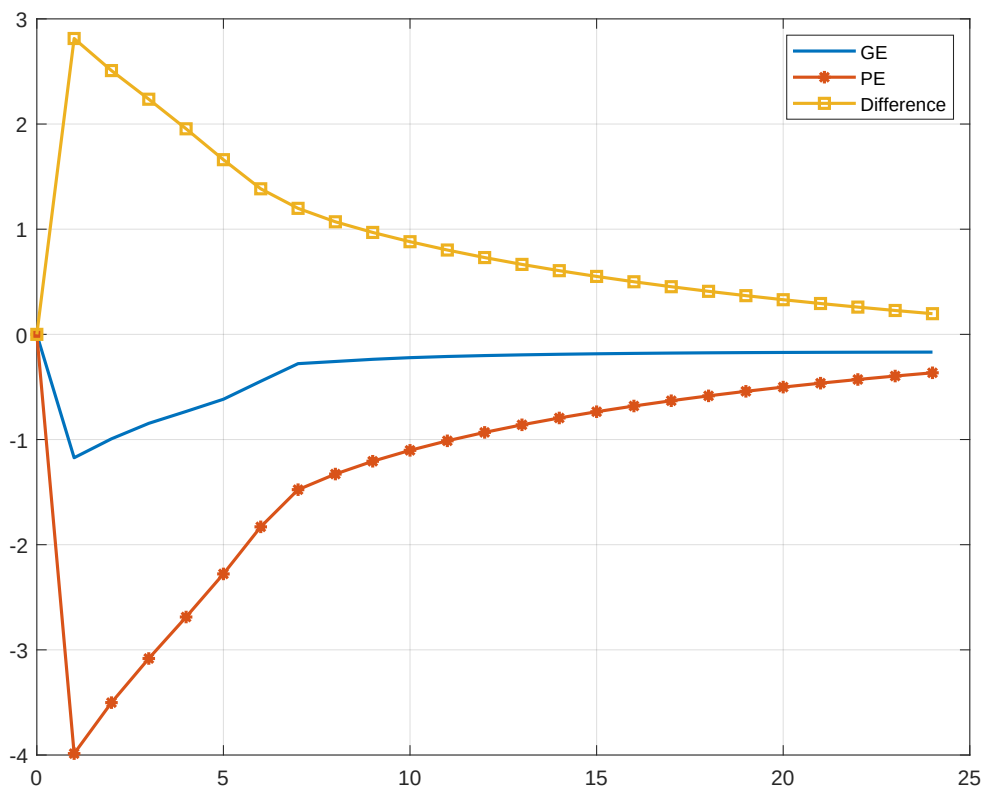


NOTES: The borrowing limit is expressed as a share of annualized total output. The interest rate is annualized and in percent. Consumption, labor supply, and output are expressed in percent deviations from their initial steady state. Vertical dotted line: period in which the borrowing constraint reaches its new steady state level.

Interest Rate Response. The interest rate experiences a sharp decline, going far below zero and reaching its trough of 1.9% after four periods. To get an intuition, it is useful to look at figure 2. As the bottom panel of figure 2 suggests, a very high fraction of households starts with very low assets in the initial steady state. After the credit tightening, part of these households will find themselves against the constraint, and will be forced to deleverage. The rest will strive to stay away from the new tighter constraint, building a stock of savings. The net effect is a strong increase in the demand for assets for low-wealth households. Given that assets are in zero net supply in equilibrium, the change in the portfolio composition of low-wealth households has to be compensated by a decrease in the supply of assets from wealthy households. Due to a combination of concave consumption and convex labor supply, wealthy households respond to the shock by running down on savings. Market clearing implies that the interest rate drops in equilibrium.

Output Response. Output contracts by 1.1% on-impact, and then it starts converging to the new lower steady state. To better understand the response of output, we should look at the partial and general equilibrium effects in both consumption and labor supply margins. Treating each margin separately and then aggregating is essential as households choose to absorb the credit shock by adjusting both consumption and labor supply. Therefore, beyond the demand side effect of consumption on output, output also relies on the supply side and the extent to which households are willing to work more to self-insure against future shocks.

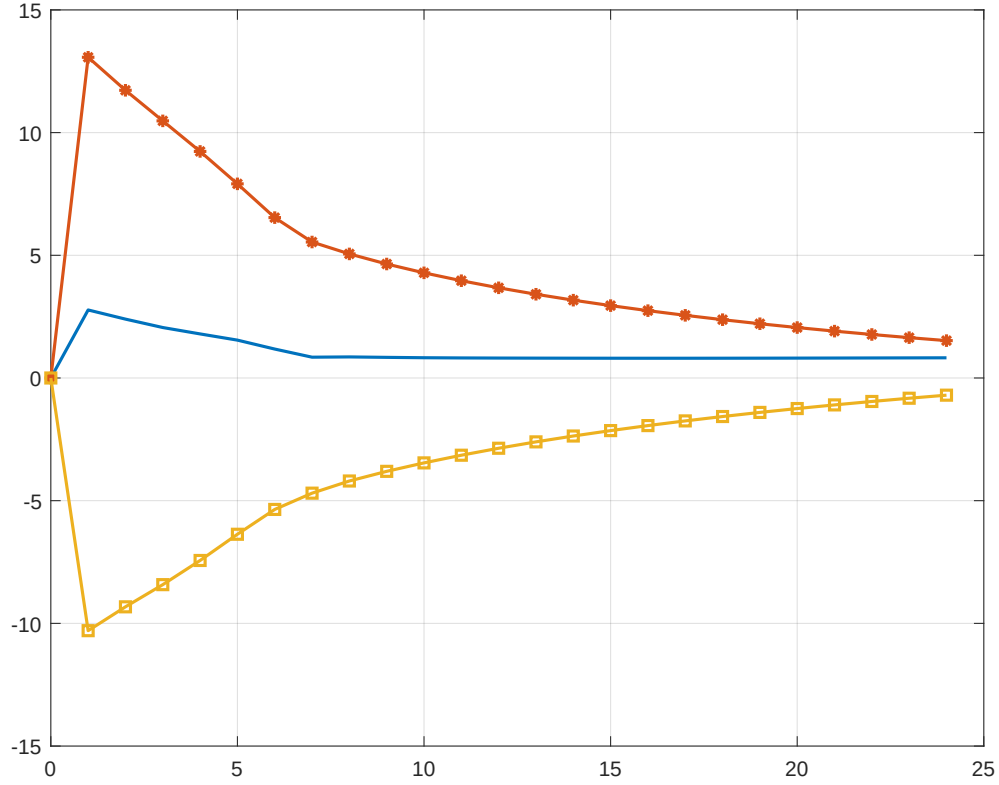
Figure 4. Consumption response decomposition | Epstein-Zin



NOTES: Solid line: General equilibrium effect. Asterisks: Partial equilibrium effect. Squares: Difference between general equilibrium and partial equilibrium effects. Consumption in percent deviations from its initial steady state value.

As we analyzed in the beginning of the subsection, the partial equilibrium effect induces both constrained and unconstrained agents to lower consumption and increase savings. Figure 4 gives a snapshot of the consumption response decomposition into partial and general equilibrium. Our simulations point towards a partial equilibrium effect that reduces consumption by 4% on impact. The same intuition applies when we look at

Figure 5. Labor supply response decomposition | Epstein-Zin

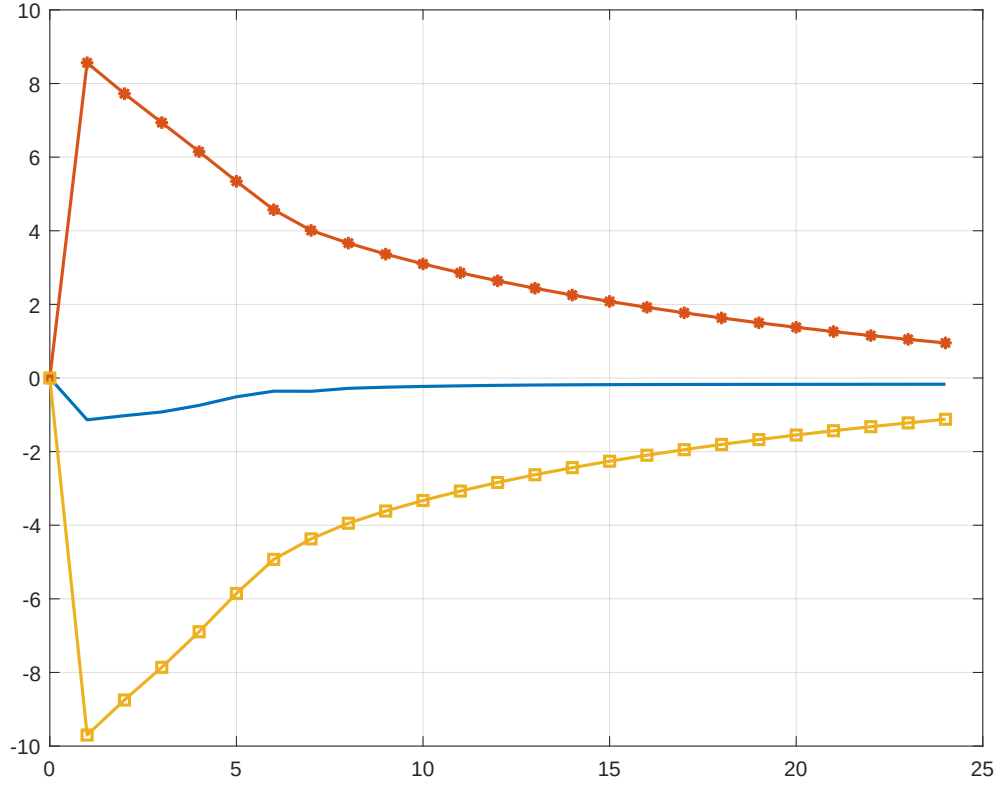


NOTES: Solid line: General equilibrium effect. Asterisks: Partial equilibrium effect. Squares: Difference between general equilibrium and partial equilibrium effects. Labor supply in percent deviations from its initial steady state value.

figure 5 and the labor margin. Initially, constrained and unconstrained households want to absorb the adverse shock partly by increasing their labor supply, which translates into a higher labor income and, consequently, higher bond holdings. We document an increase in labor supply of 13.07% on impact, in partial equilibrium.

Both effects lead to an increase in the net demand for assets. Since the supply of assets is (exogenously) fixed, the interest rate has to drop to induce equilibrium in the market for assets. As the interest rate transitions to its new lower equilibrium value, the intertemporal price of consumption increases. Moreover, leisure starts to become more attractive as its cost in terms of current consumption units decreases. Therefore, both constrained and unconstrained households adjust by increasing consumption today due to intertemporal substitution, and decreasing labor supply due to the lower cost of leisure. This is the general equilibrium effect. Quantitatively, the endogenous drop in the interest rate of 1.9% in the short-run, leads to an increase in aggregate consumption

Figure 6. Output response decomposition | Epstein-Zin



NOTES: Solid line: General equilibrium effect. Asterisks: Partial equilibrium effect. Squares: Difference between general equilibrium and partial equilibrium effects. Output in percent deviations from its initial steady state value.

of 2.8% on impact, considerably mitigating the partial equilibrium effect. The reverse intuition applies when looking at labor supply. In particular, the general equilibrium effect contracts aggregate hours worked by 10% on impact, partially offsetting the strong initial increase absent equilibrating forces.

Given our baseline calibration, the partial equilibrium effect dominates along the supply and demand sides of the economy. This points toward a reduction in aggregate consumption and an increase in aggregate labor supply. One would be tempted to infer that these two offsetting forces would have opposing effects on output. In our model this is not the case. In particular, the increased aggregate labor supply does not translate into an increase in output. This is due to compositional effects that determine the extent to which workers with different productivity levels adjust their labor supply. Since in our model labor income is tied to productivity, low-productivity households with low asset holdings are the ones to increase labor supply aggressively since they are interest-insensitive and

the partial equilibrium effect dominates, whereas high-productivity households at the right end of the asset distribution work less hours due to a dominant general equilibrium effect. Therefore, even though the partial equilibrium effect dominates in equilibrium and hours worked increase on aggregate, the contractionary effect coming from the declining labor supply of productive, wealthy households more than offsets the increase in output due to higher hours worked of under-productive, low-wealth households. In other words, the aggregate labor productivity declines. Hence, contrary to the conventional wisdom, both effects lead to a decline in output. As figure 6 suggests, although the partial equilibrium effect calls for a strong increase in output of 8.5 % on impact, this increase is balanced by the endogenous drop in the interest rate. Overall, taking into account the demand and supply side effects, output drops by 1.1% on impact.

4 INSPECTING THE MECHANISM

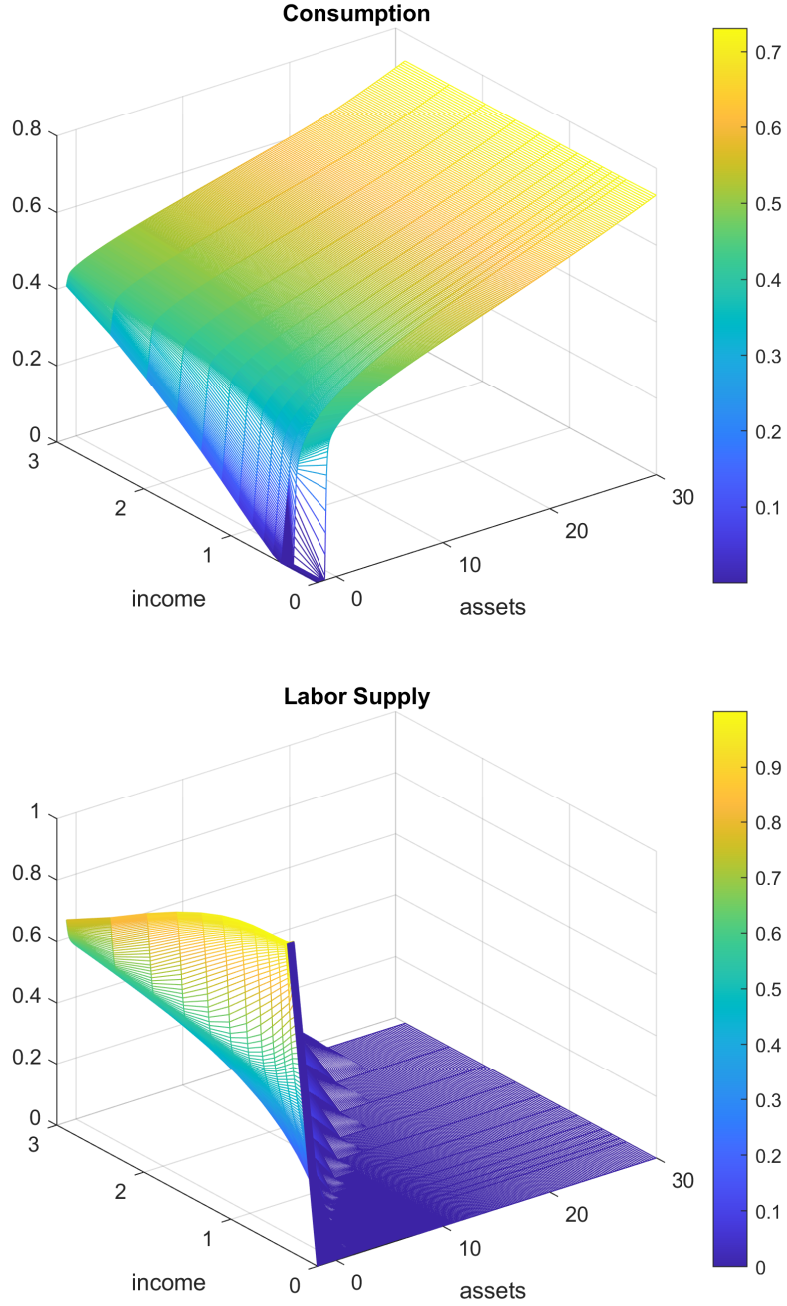
In this section we present alternative calibrations which differ with respect to the coefficient of relative risk aversion (RA), and the borrowing limits (ϕ_t), that are consistent with a debt-to-GDP ratio of 18% and 8% in the initial and terminal steady states, respectively.

4.1 CRRA vs Epstein-Zin

We begin by analyzing an economy with CRRA preferences. To this end, we set $RA = 4$, which implies an elasticity of intertemporal substitution of $1/\gamma = 1/4$. This calibration replicates the expected-utility case, as the coefficient of relative risk aversion and the elasticity of intertemporal substitution are reciprocal of one another. The main motive behind this exercise is to use the expected-utility case as a natural benchmark against which we compare our findings under Epstein-Zin preferences. This helps to uncover the extent to which the preference specification tilts the response of the economy to an unexpected one-time shock, and the implications for the business cycle.

Figure 7 reports the policy functions for consumption and labor supply when the utility function displays constant relative risk aversion (CRRA). The consumption function becomes more concave as the coefficient of relative risk aversion reduces to the value that replicates the CRRA case. This is at odds with the expected-utility benchmark, in which the concavity of consumption is a positive function of the risk aversion parameter (Guer-

Figure 7. Policy functions | CRRA

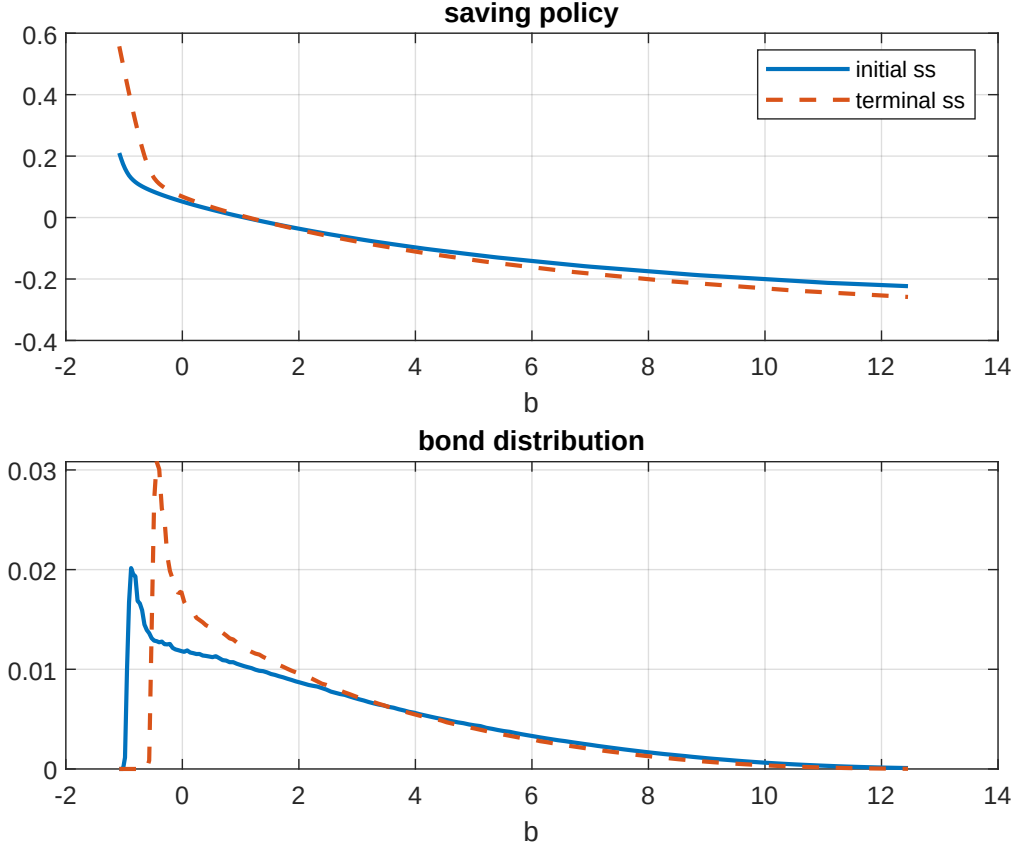


NOTES: Consumption and labor supply decision rules under CRRA preferences. Variables are expressed in levels.

[rieri and Lorenzoni \(2017\)](#)). To understand the intuition behind this result, it is useful to look at the first panel of figure 8. With CRRA preferences, the demand for savings of low wealth households is lower than when households have recursive utility. This, in turn, implies that those households have less power to self-insure against idiosyncratic income risk, facing greater consumption volatility. Greater consumption volatility is the

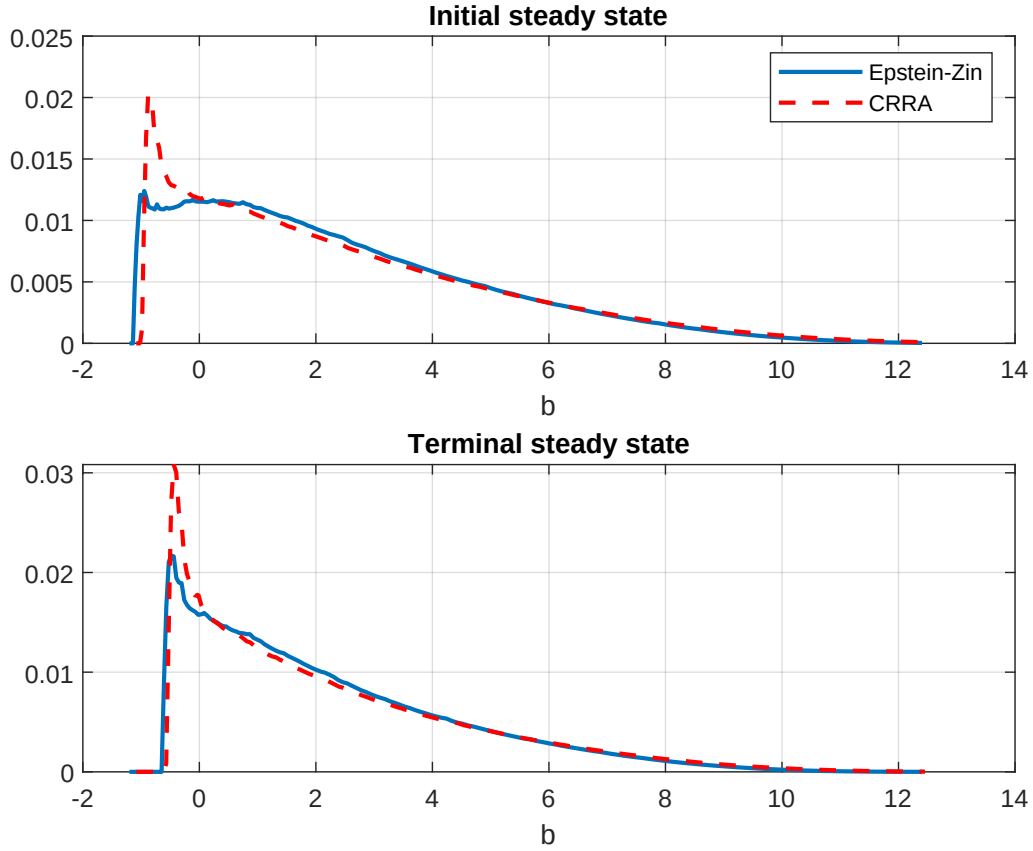
key element to account for the higher curvature of the consumption function at that region of the bond distribution.

Figure 8. Savings and bond distribution | CRRA



Beyond its impact on the shape of the consumption function, the extend to which households adjust the demand for savings has strong implications for the distribution of wealth. The panels in figure 9 contrast the density of bond holdings in the initial (top panel) and terminal (bottom panel) steady states, respectively. It is straightforward that the asset distribution in the CRRA case displays a higher mass of agents closer to the borrowing limit, relative to the distribution under recursive utility. The latter is more concentrated to higher values of asset holdings. The intuition is simple. Even absent the aggregate shock, as the degree of risk aversion decreases, households, irrespective of their initial asset position, become less prudent and start to decumulate savings. Thus, the distribution displays more households near the borrowing constraint. This, however, does not alter the aggregate amount of savings which is (exogenously) fixed and identical across preference specifications.

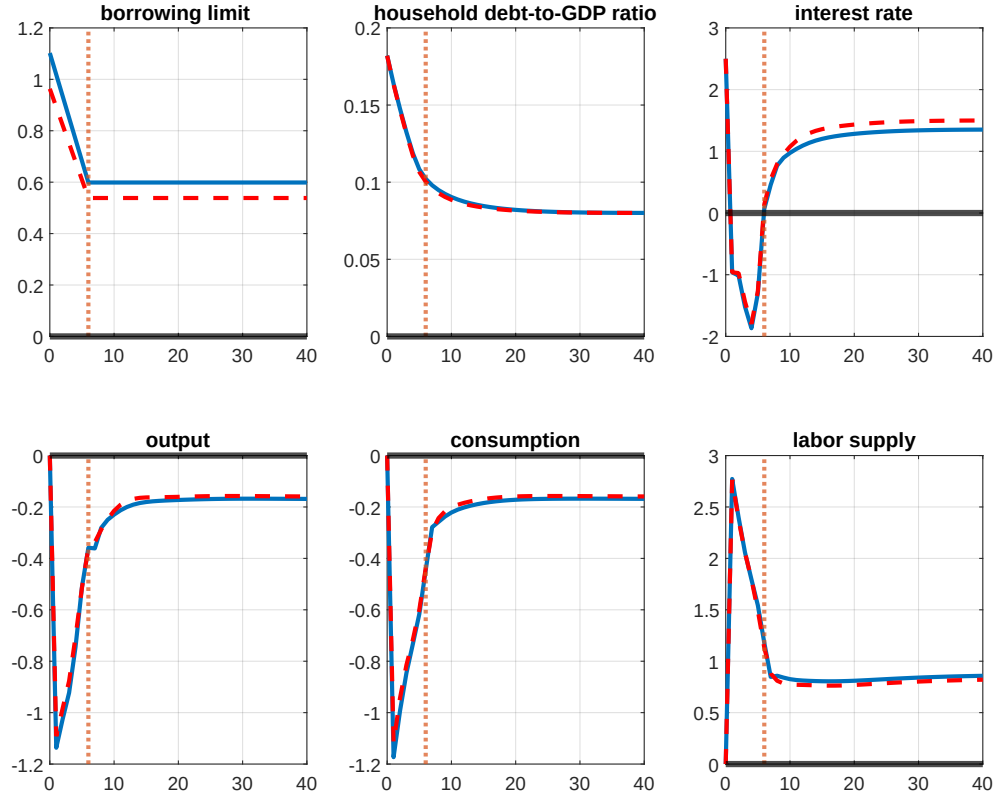
Figure 9. Bond distribution | Uniform debt/GDP targets



A common feature of the bond distribution across preference specifications is that it shifts to the right in the terminal steady state. The main idea behind the more concentrated terminal-steady-state distribution is that after the shock is realized, constrained agents are forced to adjust their consumption downwards, thus increasing their savings, while unconstrained agents engage in bond accumulation to preclude the possibility of hitting the constraint in a later period. Hence, all agents close to the constraint end up with a higher level of assets, and the terminal steady state shifts to the right.

The macroeconomic implications of households' decision making in response to a credit tightening are reflected in the panels of figure 10. The figure depicts the time-path of macroeconomic aggregates, and makes a quantitative comparison between the transition dynamics for the two main preference specifications. We observe that our variables of interest follow a qualitative pattern characterized by an aggressive downward adjustment of output, the debt-to-GDP ratio, and the interest rate, on impact. This is the outcome of the tighter borrowing limit, which depresses aggregate activity. The intuition behind

Figure 10. Impulse response functions | Uniform debt/GDP targets



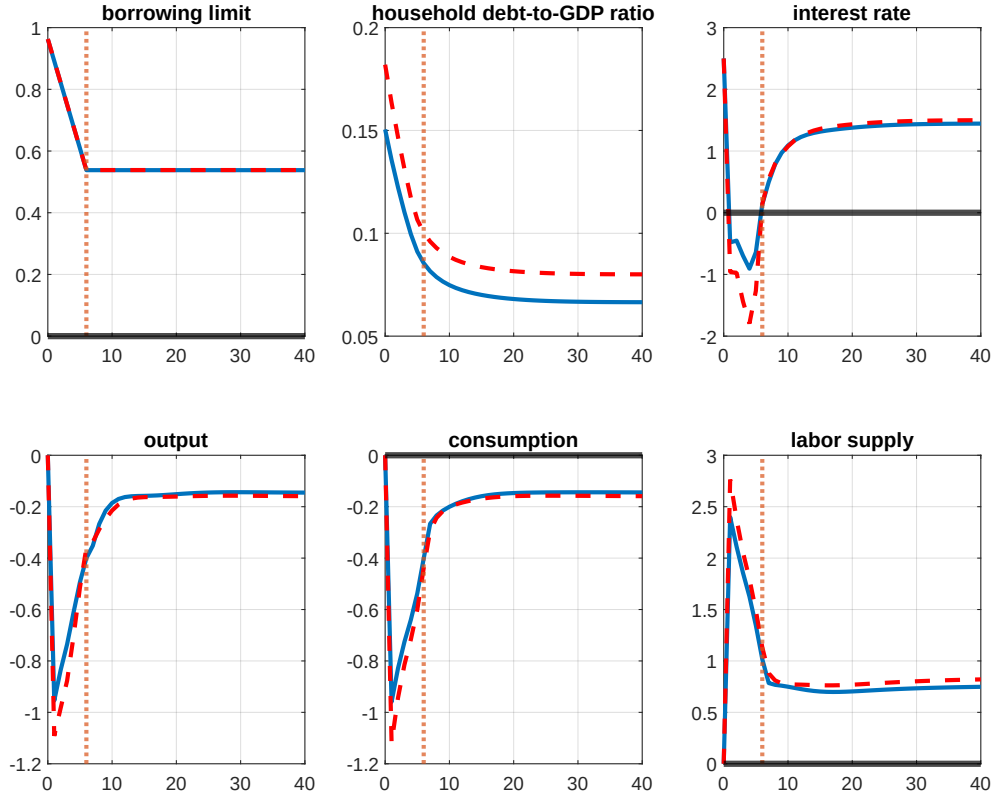
NOTES: Dashed red lines: CRRA preferences. Solid blue lines: Epstein-Zin preferences. Vertical dotted line: period in which the borrowing constraint reaches its new steady state level. The borrowing limits are expressed as a share of annual total output. The interest rates are annualized and in percent. Consumption, labor supply, and output are expressed in percent deviations from their initial steady state.

these results carries over from the Epstein-Zin case. When comparing the quantitative behavior of macroeconomic aggregates, the key takeaway is an overall similar degree of adjustment that persists for almost the entire time-horizon, with negligible differences in the terminal steady state. This suggests that our model with Epstein-Zin preferences does a fairly decent job at explaining the dynamics of the U.S business cycle in the apex of the 2008 financial crisis, producing moments that match those of the expected-utility counterpart. This is particularly encouraging for models with recursive utility calibrated under plausible parameter values, that is, parameter values within the well-respected range the macro-finance literature dictates.

4.2 Fixed borrowing limits

This section studies the quantitative implications of equalizing the borrowing limits across the main preference specifications. To this end, we take our benchmark model with Epstein-Zin preferences and fix values for the borrowing limits in initial and terminal steady states that conform to the calibrated values under the expected-utility framework. In particular, we set the borrowing constraints to $(\phi/4Y) = 0.963$ and $(\phi'/4Y) = 0.540$,

Figure 11. Impulse response functions | Uniform borrowing constraints

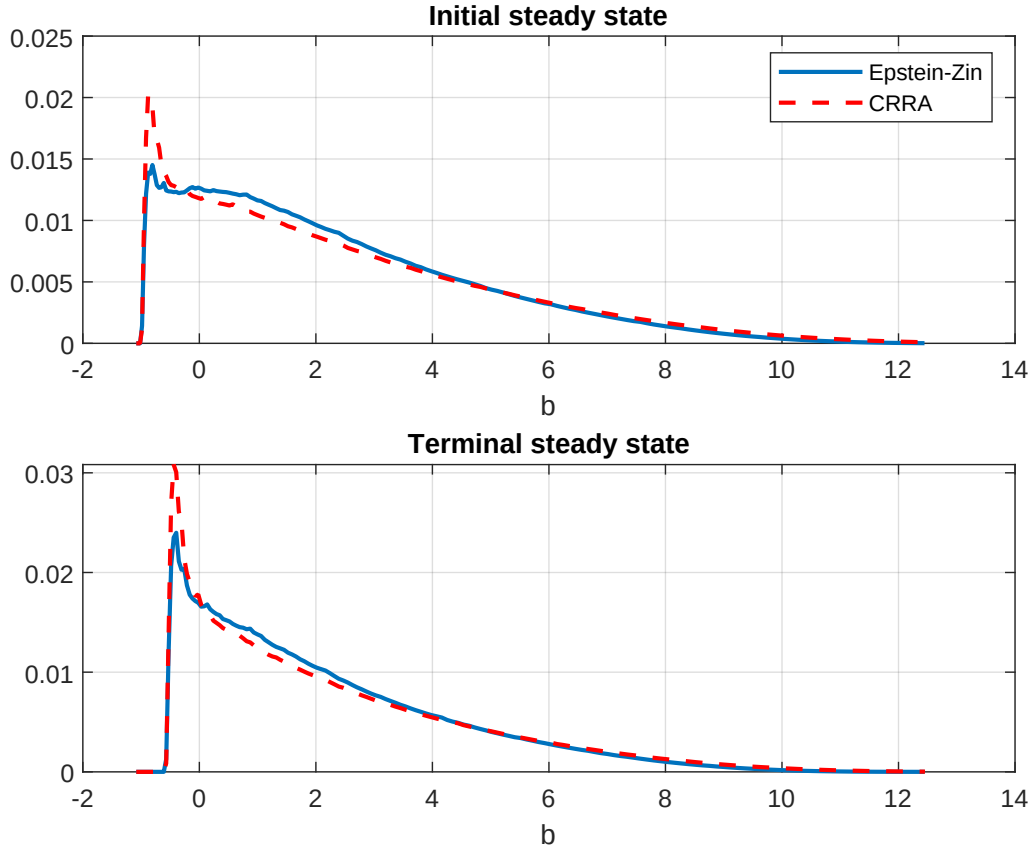


NOTES: Dashed red lines: CRRA preferences. Solid blue lines: Epstein-Zin preferences. Vertical dotted line: period in which the borrowing constraint reaches its new steady state level. Borrowing limits are expressed as a share of annual total output. The interest rate is annualized. Output, consumption, and labor supply in percent deviations from their initial steady state values.

respectively. The implication of this tweak is that we no longer target a specific debt-to-GDP ratio before and after the shock; rather we let it adjust to levels consistent with ϕ and ϕ' . The bottom line is that we can study the extent to which the behavior of macro aggregates in both economies differ when hit by a shock of identical size.

Figure 11 shows the full transition path of the main macro aggregates following a credit tightening. The key takeaway from the impulse response analysis is an overall smoother

Figure 12. Bond distribution | Uniform borrowing constraints



adjustment to the shock in the Epstein-Zin framework. The reason behind this result is straightforward. With Epstein-Zin preferences - and the borrowing limits fixed to the values prevailing in the CRRA case - the required degree of downward adjustment in the debt-to-GDP ratio is lower. In particular, the debt-to-GDP ratio falls by roughly 8 percentage points in the terminal steady state compared to the 10 percentage point decline when households have time-separable preferences. The reasoning behind the softer landing follows from the structure of Epstein-Zin utility, and in particular the common practice of endowing households with greater degrees of risk aversion. Higher risk aversion in turn makes the marginal utility more convex, and households more prudent. This implies that households issue less debt in the initial steady state, attenuating the spillovers from a tighter borrowing constraint.

Figure 12 shows the ergodic distribution of bond holdings in the initial and terminal steady states across preference specifications. The top panel shows that when preferences follow the Epstein-Zin recursive structure, a lower fraction of households is concentrated

around the borrowing limit before the shock. Less excessive (initial) steady state leverage implies a less costly adjustment to the tighter borrowing limit. Therefore, consumption and output fall by less, and labor supply increases by less. Consistent with the smoother downward adjustment of the stock of savings, the response of the real rate is attenuated.

5 MODEL WITH CREDIT SPREADS

In this section we extend the baseline model by relaxing the simplifying assumption of a single interest rate, allowing for a wedge between lending and deposit rates. Similar to [Kaplan et al. \(2018\)](#), households can hold negative liquid assets b bounded by an exogenous debt limit ϕ , at the real interest rate $r^l = r^d + \kappa$. Parameter $\kappa \in (0, 1)$ reflects the spread between the rate charged by banks on loans to the private sector (r^l), and the rate paid by the same institutions on short-term deposit contracts (r^d). To aid exposition, we take κ as exogenous.

To motivate the spread between the lending and deposit rates, we can think of κ as a transaction fee charged by banks to intermediate funds from lenders to borrowers in a perfectly competitive market. If household i is a net lender ($b'_i > 0$), the household purchases bonds at a price equal to r^d . If the household is a net borrower ($b'_i < 0$), the household issues non-contingent debt by selling bonds at a price $r^d + \kappa$. The difference between the lending and deposit rates, κ , captures bank transaction costs when providing intermediation services. These intermediation costs may reflect screening and monitoring costs. We treat κ as a deadweight loss for the economy, and depart from [Boissay et al. \(2016\)](#) who assume that households are rebated the intermediation costs through lump-sum transfers. Following [Guerrieri and Lorenzoni \(2017\)](#), we assume that these costs are linear and equal to κ for every unit of traded bonds, and banks are subject to a zero-profit constraint.

If we let positive and negative bond holdings be denoted by the respective b^+ and b^- , the household budget constraint can be expressed as

$$\underbrace{qb'_i}_{+} + \underbrace{\hat{q}b'_i}_{-} + c_i \leq b_i + y_i - \tilde{\tau}_i, \quad (14)$$

where $\hat{q} = (1 - \kappa)q$ is the price net borrowers receive for each unit of bonds issued.

Term $(+)$ constitutes the total amount net lenders pay to purchase bonds. Term $(-)$ stands for the total amount net borrowers receive when issuing bonds. The amount net borrowers receive is clearly lower than the amount net lenders pay, with the difference being absorbed by the bank to compensate for the costly provision of intermediation services. The rest of the model is identical to the baseline version outlined in [Section 2](#).

5.1 Higher credit spreads

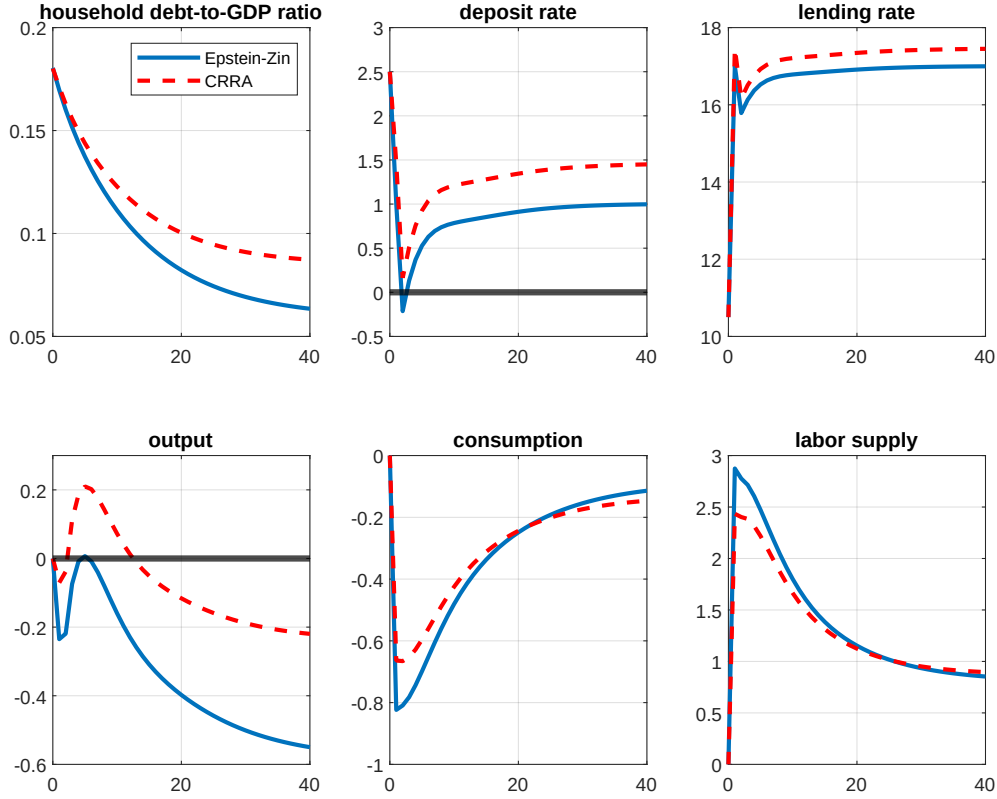
We perform an exercise that aims to track the response of the economy to a different type of a credit tightening, namely a widening of credit spreads. This is similar in nature to the tightening of the borrowing constraint, in that both induce households to deleverage and improve their asset positions. The idea in this case is that higher lending rates elevate the cost of borrowing, prompting households to demand less credit.

To do so, we shock the economy with a sudden and unexpected permanent increase in the spread κ . The shock is exogenous and affects all households symmetrically. Quantitatively, we set κ to two percent in the initial steady state, and let it jump discretely by two percentage points, at a quarterly frequency.

[Figure 13](#) shows the behavior of key aggregate variables along the perfect foresight transition to the new steady state. In period $t = 0$ the economy sits in rest, and in $t = 1$ there is a sudden and unexpected permanent spike in the intermediation cost of two percentage points (quarterly) that drives up the credit spread. As the spread widens, the cost of borrowing rises, inducing households to demand less credit. Consumption shrinks, triggering a subsequent fall in the debt/GDP ratio and output. This creates an imbalance in the asset market, as the fixed supply of bonds can not be met by the higher asset demand coming from households' desire to improve their debt positions. The deposit rate drops to discourage savings and re-establish equilibrium in the bond market. Following the consumption drop, an additional margin of adjustment is activated through the endogenous response of labor supply. In this regard, households partly absorb the shock by increasing the hours worked to partially offset the negative effects on aggregate demand and sustain a smooth consumption path.

Differentiating across preference specifications, [figure 13](#) suggests an amplification of the pass-through to aggregate variables in the Epstein-Zin economy relative to the case with expected utility. This is because, even though the shock is uniform and symmetric

Figure 13. Impulse response functions | Uniform spike in credit spreads



NOTES: Permanent spread shock. Dashed red lines: CRRA preferences. Solid blue lines: Epstein-Zin preferences. Deposit and lending rates are annualized and in percent. Output, consumption, and labor supply in percent deviations from their initial steady state values.

to all households, the economy with recursive utility exhibits a significantly stronger downward adjustment in the debt/GDP ratio in the terminal steady state. Specifically, with Epstein-Zin utility the borrowing constraint consistent with a 18% initial debt/GDP ratio is looser. As Epstein-Zin households by construction are more risk averse, to induce similar initial debt/GDP levels across preferences we need a higher mass of borrowing households in the Epstein-Zin economy. Thus, the Epstein-Zin economy features a more concentrated bond distribution close to the constraint in the initial steady state. Because with the spread shock all the adjustment comes from the extensive margin, this implies that in the model with recursive utility a larger mass of households is affected when κ increases. Hence, more households engage in asset accumulation (i.e. deleverage), and the aggregate responses get amplified.

6 CONCLUSION

In this paper we add Epstein-Zin preferences in an otherwise standard incomplete markets model with idiosyncratic and aggregate shocks. The reason for choosing this setup is twofold. First, it serves as a benchmark to understand the extent to which altering the preference specification has a meaningful quantitative impact on households' decision-making – and the macro aggregates — when the borrowing constraint tightens unexpectedly. Second, recursive utility provides an extra degree of freedom as it disentangles timing from risk attitudes, providing a better approximation of asset returns and remaining free from the renowned puzzles in the asset pricing literature.

Quantitatively, the setup with Epstein-Zin preferences displays a smoother adjustment of macro aggregates compared to a model with CRRA utility. This is due to the fact that the required degree of downward adjustment in the debt-to-GDP ratio is lower, since households become more prudent and accumulate more savings in the initial steady state. These predictions are reversed when we introduce a spread between the borrowing and lending rates, such that the economy with recursive utility amplifies the response of macro aggregates to a spread hike.

The model can be extended in many different directions if we want to get richer dynamics and capture other interesting dimensions of the economy. First, adding risky assets is a natural extension of our Epstein-Zin specification, as it allows to study both the behavior of asset prices and the macroeconomic impact of shocks in a unified framework. Second, adding capital accumulation in a framework in which long-run growth is subject to firms' investment in innovation can give rise to endogenous business cycles ([Benigno and Fornaro, 2018](#)). This is useful if we want to study whether alternative preference specifications can have real effects on long-run growth prospects. Finally, introducing nominal rigidities and exploring a mix of fiscal and monetary policies in a novel setup is always of interest.

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Appendix

A Optimality Conditions

To get the Euler equations for consumption and leisure we take the following steps. First, we get the first order condition with respect to consumption, which can be written as

$$\begin{aligned}
 u_c(c, n) + \beta \frac{1}{1-\alpha} (EV(b', \theta')^{1-\alpha})^{\frac{\alpha}{1-\alpha}} (1-\alpha) E (V(b', \theta')^{-\alpha} V_c(b', \theta')) &\geq 0 \\
 u_c(c, n) &\geq -\beta (EV(b', \theta')^{1-\alpha})^{\frac{\alpha}{1-\alpha}} E \left(V(b', \theta')^{-\alpha} V_{b'}(b', \theta') \left(\frac{-1}{q} \right) \right) \\
 u_c(c, n) &\geq \frac{\beta}{q} [EV(b', \theta')^{1-\alpha}]^{\frac{\alpha}{1-\alpha}} E [V(b', \theta')^{-\alpha} V_{b'}(b', \theta')], \quad (A1)
 \end{aligned}$$

and the optimality condition with respect to labor supply

$$\begin{aligned}
 u_n(c, n) + \beta \frac{1}{1-\alpha} (EV(b', \theta')^{1-\alpha})^{\frac{\alpha}{1-\alpha}} (1-\alpha) E \left(V(b', \theta')^{-\alpha} V_{b'}(b', \theta') \frac{w}{q} \right) &\leq 0 \\
 u_n(c, n) &\leq -\frac{\beta}{q} w [EV(b', \theta')^{1-\alpha}]^{\frac{\alpha}{1-\alpha}} E [V(b', \theta')^{-\alpha} V_{b'}(b', \theta')]. \quad (A2)
 \end{aligned}$$

The problem that pops up is that we do not know how to find the derivative of the value function $V(b', \theta')$ with respect to b' . However, if we assume that b' has been chosen optimally, we can take the derivative of the value function with respect to b , given by

$$\begin{aligned}
 \frac{\partial V(b, \theta)}{\partial b} &= u_c(c, n) \frac{\partial c}{\partial b} + u_n(c, n) \frac{\partial n}{\partial b} + \frac{\beta}{1-\alpha} [EV(b', \theta')]^{\frac{\alpha}{1-\alpha}} (1-\alpha) E \left(V(b', \theta')^{-\alpha} \frac{\partial V(b', \theta')}{\partial b'} \frac{\partial b'}{\partial b} \right) \\
 &= u_c(c, n) \frac{\partial c}{\partial b} + u_n(c_t, n_t) \frac{\partial n}{\partial b} + \frac{\beta}{1-\alpha} (EV(b', \theta'))^{\frac{\alpha}{1-\alpha}} (1-\alpha) \\
 &\quad \times E \left(V(b', \theta')^{-\alpha} \times \frac{\partial V(b', \theta')}{\partial b'} \left(\frac{1}{q} + \frac{w}{q} \frac{\partial n}{\partial b} - \frac{1}{q} \frac{\partial c}{\partial b} \right) \right). \quad (A3)
 \end{aligned}$$

Combined together, we get the Benveniste-Scheinkman envelope condition

$$\begin{aligned}
\frac{\partial V(b, \theta)}{\partial b} &= \frac{\beta}{q} [EV(b', \theta')^{1-\alpha}]^{\frac{\alpha}{1-\alpha}} E \left[V(b', \theta')^{-\alpha} V_{b'}(b', \theta') \frac{\partial c}{\partial b} \right] - \frac{\beta}{q} w [EV(b', \theta')^{1-\alpha}]^{\frac{\alpha}{1-\alpha}} \\
&\times E \left[V(b', \theta')^{-\alpha} V_{b'}(b', \theta') \frac{\partial n}{\partial b} \right] + \frac{\beta}{q} [EV(b', \theta')^{1-\alpha}]^{\frac{\alpha}{1-\alpha}} E [V(b', \theta')^{-\alpha} V_{b'}(b', \theta')] \\
&+ \frac{\beta}{q} w [EV(b', \theta')^{1-\alpha}]^{\frac{\alpha}{1-\alpha}} E \left[V(b', \theta')^{-\alpha} V_{b'}(b', \theta') \frac{\partial n}{\partial b} \right] - \frac{\beta}{q} [EV(b', \theta')^{1-\alpha}]^{\frac{\alpha}{1-\alpha}} \\
&\times E \left[V(b', \theta')^{-\alpha} V_{b'}(b', \theta') \frac{\partial c}{\partial b} \right] \\
&= \frac{\beta}{q} [EV(b', \theta')^{1-\alpha}]^{\frac{\alpha}{1-\alpha}} E [V(b', \theta')^{-\alpha} V_{b'}(b', \theta')]. \tag{A4}
\end{aligned}$$

Now, combining (A1) and the envelope condition (A4), we get the Euler for consumption

$$\begin{aligned}
u_c(c, n) &\geq \frac{\beta}{q} [EV(b', \theta')^{1-\alpha}]^{\frac{\alpha}{1-\alpha}} E [V(b', \theta')^{-\alpha} u_{c'}(c', n')] \\
u_c(c, n) &\geq \beta(1+r) [EV(b', \theta')^{1-\alpha}]^{\frac{\alpha}{1-\alpha}} E [V(b', \theta')^{-\alpha} u_{c'}(c', n')]. \tag{A5}
\end{aligned}$$

This is an intertemporal optimality condition, and holds with equality when the exogenous debt limit $b' \geq 0$ is not binding. Finally, combining equations (A1) and (A2), we get the Euler for labor supply

$$\begin{aligned}
u_n(c, n) &\leq -w u_c(c, n) \\
-\frac{u_n(c, n)}{u_c(c, n)} &\geq w. \tag{A6}
\end{aligned}$$

This is an intratemporal optimality condition that equates the marginal rate of substitution between consumption and leisure with the real wage. It holds with equality when $n \geq 0$.

A Solution Algorithm

To compute the optimal decision rules for consumption and labor supply, we proceed by discretizing the state space. We setup a non-linear grid for the endogenous state (assets) comprising 300 values. The grid is denser for lower values of bonds, because the policy functions at that region display higher curvature, and hence more gridpoints are

needed to get a better approximation. We fix the upper bound of the asset grid at 50, and the lower bound at -3. Since the decision rules are functions of the asset choices and the income realizations, they are allowed to be nonlinear between gridpoints.

The employment status follows an AR(1) stochastic process which is discretized using 12 equally spaced nodes, delivering a probability transition matrix of size 12x12. For the discretization we use the method of [Tauchen \(1986\)](#).

To solve for the equilibrium, we employ the method of Endogenous Gridpoints originally proposed by [Carroll \(2006\)](#). The main advantage of this method is that it skips the optimization step necessary to build the Euler equation, the most time-consuming step in the traditional Value Function Iteration approach.

The complexity of the model increases when we introduce an aggregate shock in borrowing capacity, on top of the idiosyncratic income shocks that hit households each period. This is because, aggregate uncertainty introduces an extra state variable that agents need to take into account when making optimal decisions about consumption and labor supply, the wealth distribution. This can be a high dimensional, or, sometimes, an infinite dimensional object, making the analysis very challenging. Most of the literature on incomplete markets models with aggregate uncertainty relies on computationally expensive techniques, such as the one developed in [Krusell and Smith \(1998\)](#), which approximates the wealth distribution with a small number of moments that incorporate all the necessary information about that distribution. In our model, we assume away aggregate uncertainty by focusing on an MIT shock, after which agents have perfect foresight. Hence, the distribution of assets does not enter the households' state space and needs not be kept track of, reducing the number of the relevant state variables and problem's dimensionality.

To compute the ergodic set of bonds, we start from a uniform distribution that reflects the percentage (fraction) of agents at each employment-asset node, and iterate on the probability transition matrix until convergence. In this way, we keep track of the fraction of agents at each node across time, and not of individual agents. In principle, we could have proceeded by finding the eigenvector of the probability transition matrix consistent with the unitary eigenvalue, but since the matrix is big this would be more costly in terms of computational time.

Finally, to compute the transitional dynamics we proceed by backwards induction.

In particular, we start by giving a guess about the full path of interest rates. Then, we assume that the economy is in the terminal steady state in period $t = T$ (we set $T = 100$) with $r_t = r_T$, and iterate on the Euler equation backwards, using the endogenous gridpoints method, to get the individual policy rules from $t = T - 1$ to $t = 0$. Next, we iterate on the bond distribution forward from $t = 0$ to $t = T$, using the computed policy rules, to calculate the aggregate demand for bonds at each period. Finally, we check for market clearing; if the asset market does not clear, we update the interest rate using the simple linear rule $r_t^\kappa = r_t^{(\kappa-1)} - w \cdot (B^s - B^d)$, where κ denotes the κ -th iteration, and start over again; otherwise we stop.

A Elasticity of Intertemporal Substitution

Figure 14 displays the policy functions when we increase the elasticity of intertemporal substitution to 0.5 (hence, γ equals 2 in Table 1), keeping the parameter of relative risk aversion fixed at 15. The effect of a higher EIS on consumption is conditional on agents' wealth position in the distribution. For wealthy households, the increase in the EIS makes current consumption more attractive, raising the demand for goods at the cost of lower consumption in the future. The opposite is true for households with levels of wealth below a threshold, with higher EIS be associated with intertemporal substitution in consumption, hence higher consumption in subsequent periods. At the aggregate level, the increased demand for consumption of wealthy households dominates the subdued demand of households with asset levels below the threshold, and consumption increases.

Besides the effect on consumption, raising the elasticity of intertemporal substitution has a direct effect on the steady state level of savings (figure 15). In particular, when we move from $\gamma = 4$ to $\gamma = 2$, the steady-state level of savings jumps only by 4.9%, indicating that the elasticity of intertemporal substitution has a minor contribution to savings decisions. These results accord with the findings in Wang et al. (2016).

Figure 14. Policy functions | $\gamma = 2$ | Epstein-Zin

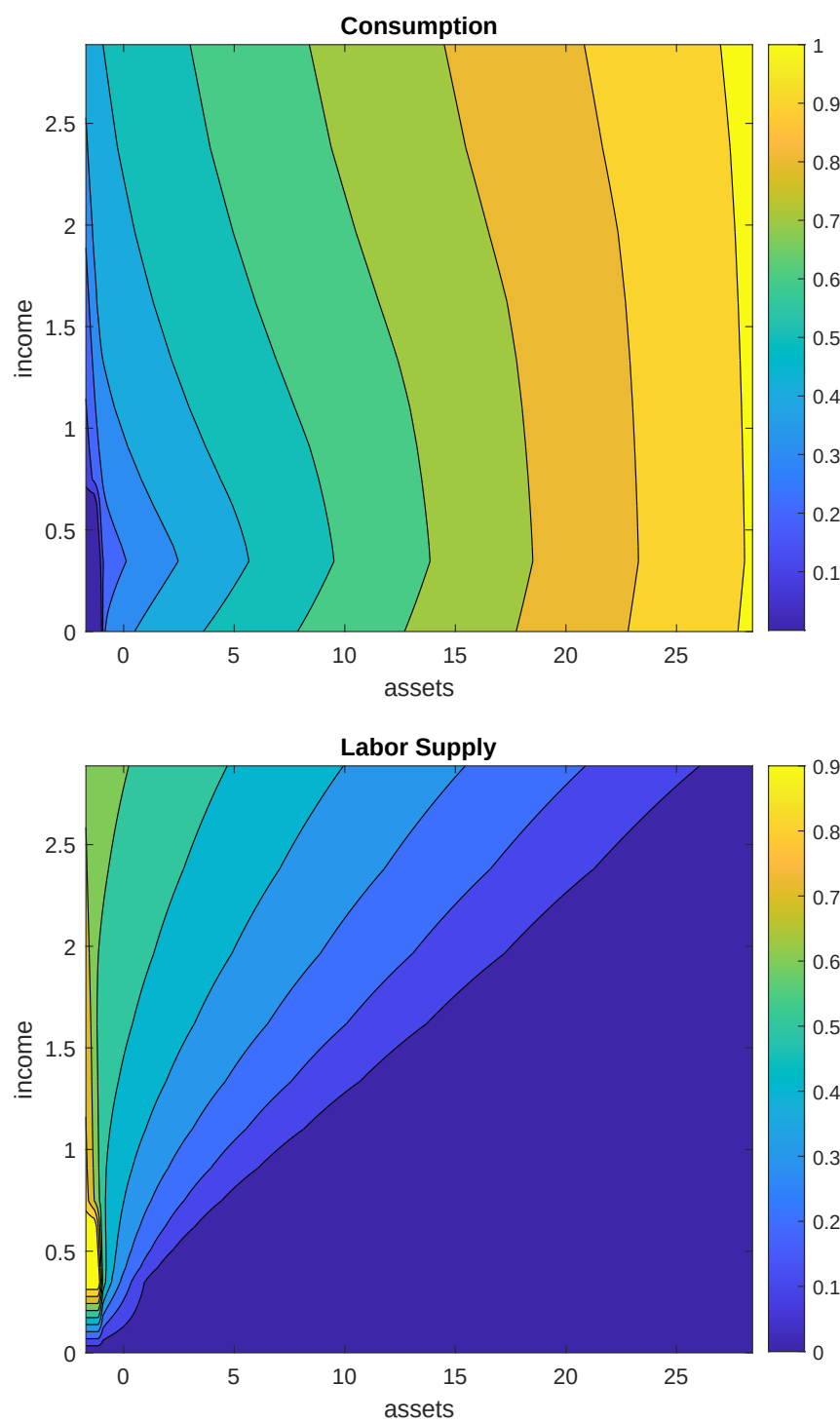
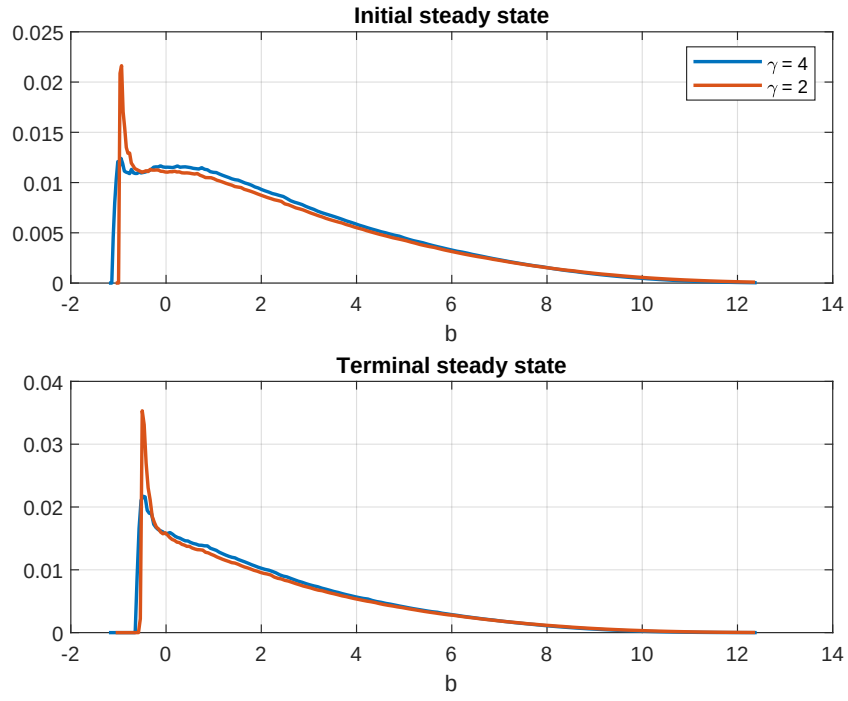
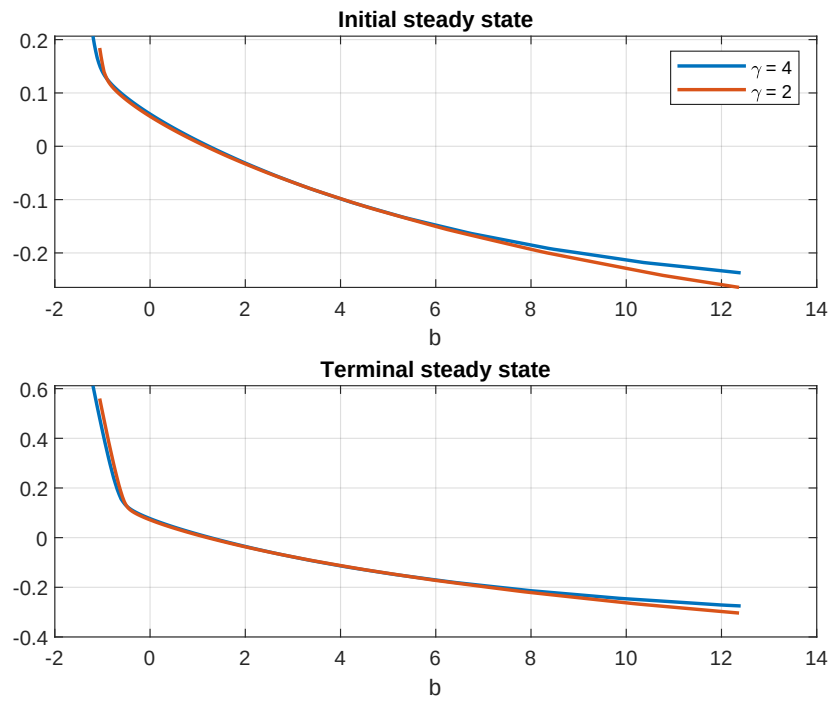


Figure 15. Savings and bond distribution | $\gamma = 2$ & $\gamma = 4$ | Epstein-Zin



(a) Bond distribution



(b) Savings